Resolving the Cosmological Scaling Crisis via a Bidirectional Fractional Cascade on a Riemannian Number Torus

Theoretical Physics Framework

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Abstract

The divergence between early-universe and late-universe measurements of the Hubble constant (H_0), known as the Hubble tension, remains one of the most significant crises in modern cosmology. This paper proposes a novel geometric and causal resolution by abandoning the assumption of a flat, linear number field for physical dimensions. Instead, we model space-time as an intrinsically curved, 2π-periodic Riemann number torus existing on the 3-sphere event horizon of a collapsing four-dimensional (4D) star. In this framework, the speed of light c is the fundamental resonance frequency of the toroidal vacuum, dynamically coupled to the cosmic expansion via the fine-structure constant α. We show that cosmological expansion is driven by a bidirectional cascade, originating from a collapsed 2D progenitor and fueled by 4D mass accretion. The Hubble tension dissolves as a geometric projection artifact. Finally, we derive the inescapable fate of the universe: an asymptotic vacuum collapse at the critical threshold $\alpha \to 1$, triggering a cataclysmic energy feedback loop that detonates the host 4D black hole.

1 Introduction & Mathematical Foundation

Standard ΛCDM cosmology relies on the implicit axiom that the underlying mathematical operators $(+, -, \times, /)$ map onto an uncurved, infinite number line $\mathbb{R}$, utilizing standard Euclidean arithmetic. However, General Relativity teaches that space-time is intrinsically curved. We extend this principle to the numerical domain of the operators themselves, defining physical space-time coordinates as existing on an intrinsically curved, 2π-periodic **Riemannian Number Torus**.

Historically, Albert Einstein proposed a closed, finite 3-sphere universe in 1917 [1]. Concurrently, Bernhard Riemann's function theory utilized multi-sheeted surfaces to map complex, non-linear dependencies without singularities [2]. Combining these paradigms, we define the current universe as a specific sheet of a multi-dimensional Riemannian manifold, where the speed of light c acts as the branch point (*Verzweigungspunkt*) or topological limit, rather than a kinetic barrier.

2 Contextual Urgency: JWST Anomalies and the Crisis of ΛCDM

The traditional cosmological paradigm, the ΛCDM model, has encountered severe empirical challenges due to recent data from the James Webb Space Telescope (JWST). JWST has revealed the existence of highly luminous, massive galaxies at redshifts of $z > 10$—an epoch where standard hierarchical structure formation dictates such mature cosmic bodies should not exist.

To reconcile these "too-early, too-massive" galaxies without invoking exotic, fine-tuned physics, modern theoretical cosmology has begun exploring two major avenues in the last 12 months:

1. **Variable Speed of Light (VSL) Frameworks:** Adjusting the value of c in the early universe shifts the look-back time and cosmic expansion rate, effectively re-calibrating the age of early structures.

2. **Topological and Closed Universes:** Exploring non-trivial topologies, such as 3-tori or closed 3-spheres, to alter the global geometry and eliminate the flat-space horizon problem.

While these contemporary approaches successfully mitigate individual anomalies, they treat

VSL and non-trivial topologies as ad-hoc modifications to an otherwise Euclidean background. This is precisely where our framework innovates: We demonstrate that a variable speed of light $c(t)$ and a closed toroidal topology are not independent cosmological phenomena, but the unified, inevitable consequence of a non-linear, Riemannian arithmetic. By treating the number field itself as a curved 2π-periodic torus, the apparent early maturity of JWST galaxies and the Hubble tension dissolve simultaneously into a singular geometric property of the space-time manifold.

3 The Unified Dimensional Cascade: 2D Spatial Genesis and 4D Accretion Mechanics

To bypass the infinite regress of top-down bulk cosmologies (i.e., the structural origin of the higher-dimensional bulk itself), we introduce a bidirectional **Fractal Dimensional Cascade**. Black holes are established as universal topological gateways operating across any $2 + n$ dimensional manifold through systematic dimensional reduction.

3.1 Spatial Genesis: The Primordial 2D Progenitor

Consider a baseline 2D quantum manifold where information density σ accumulates over cosmic cycles. Upon reaching the critical Bekenstein saturation bound, the linear 2D arithmetic collapses. To conserve information (Uni-Unitarity), the 2D sheet undergoes a global topological folding mechanism:

$$\lim_{\sigma \to \sigma_{\text{crit}}} \text{Area}_{2D} \implies \text{Boundary}(\text{Volume}_{3D}) \quad (1)$$

This geometric phase transition dynamically generates the third spatial dimension. Deep-space cartography of our large-scale structure—specifically the sharp, sheet-like *Filaments* enclosing vast, underdense *Voids* (the Cosmic Web)—serves as the un-erased, archetypal birth scar of this original 2D manifold's collapse.

3.2 Symmetric Dimensional Reduction: From 2D Discs to 3D Spheres

A profound geometric symmetry governs the collapse of matter into topological singularities across adjacent dimensional boundaries. In a standard 3D space, gravitational collapse causes a radial dimension (r) to freeze at the Schwarzschild radius, mapping the event horizon onto a 2D spherical surface (S^2). Due to the conservation of angular momentum, infalling matter is forced into an orthogonal plane, collapsing a single spatial dimension to form a highly flattened, **2D accretion disc**.

By direct dimensional analogy, the collapse of a 4D stellar progenitor within a 4D bulk undergoes an identical reduction: the collapse of the fourth spatial coordinate (w) maps the 4D event horizon onto a **3D hyperspherical surface** (S^3)—representing our physical observable universe. Concurrently, the angular momentum of the infalling 4D bulk matter undergoes a single-dimension collapse, reorganizing the influx not into a flat disc, but into an isotropic, rotating **3D accretion sphere**.

3.3 Causal Dynamics and Stochastic Winding Frequencies

Our universe is thus embedded directly within the active 3D accretion sphere of a host 4D black hole. The ongoing expansion of our spatial metric is driven by the continuous top-down influx of higher-dimensional matter-energy. Holographically delocalized across the 3-sphere boundary, this input manifests uniformly as the cosmic fundamental wave (*Grundwelle*) that sustains the quantum vacuum frequency f_0.

Mathematically, the radius of the hyperspherical horizon $R(t)$ scales directly with the accumulated 4D mass-energy $M_{4\text{D}}(t)$, deriving the Hubble expansion rate H_0:

$$R(t) = \gamma \cdot \sqrt{M_{4\text{D}}(t)} \quad \implies \quad H_0 \equiv \frac{\dot{R}}{R} = \frac{\dot{M}_{4\text{D}}}{2M_{4\text{D}}} \quad (2)$$

Crucially, the initial angular momentum (L_{4D}) of the collapsing 4D stellar progenitor is a stochastic variable governed by the turbulent, non-isotropic distribution of primordial bulk matter. Because L_{4D} dictates the initial rotation velocity and boundary conditions of the 3D accretion sphere, the fundamental vacuum resonance f_0, the baseline expansion rate H_0, and the calibrated values of our physical constants are not absolute cosmic imperatives. They are stochastically selected eigenvalues unique to our specific host black hole within a wider, multi-dimensional bulk ensemble.

4 The Toroidal Geometric Model and the Deconstruction of Linear Operators

Standard cosmological scaling laws inherently rely on the application of the natural logarithm (ln). For instance, in an idealized flat expansion, entropy, scale factor relations, and field couplings are integrated via the fundamental relation:

$$\ln(x) = \int_1^x \frac{1}{t}\, dt \tag{3}$$

However, this definition presumes a continuous, infinite, and isotropic number line $\mathbb{R}$. On a 2π-periodic Riemannian number torus, this assumption fails fundamentally due to the presence of topological constraints and branch points.

Integrating a density or distance function over a compact, closed manifold introduces global holonomy. When moving along the toroidal or poloidal cycles, the path encounters its own boundary conditions every 2π. Consequently, the flat metric collapses, and the linear integration path $\frac{1}{t}dt$ winds around the manifold, turning the simple logarithm into a multi-valued function with singular poles at the branch points. Therefore, linear arithmetic operations $(+, -, \times, /)$ and continuous transcendental functions $(\ln, \log)$ must be discarded for physical dimensions.

To preserve the conservation of information (Uni-Unitarity) without relying on flat-space integration, the logarithm is replaced by a **strict projective modular ratio**. Let the universe be a Clifford torus embedded within the 3-sphere horizon, characterized by two distinct cyclic paths: the major toroidal radius R, representing the macroscopic scale of the observable universe, and the minor poloidal radius r, representing the microscopic quantum-vacuum thickness.

Instead of a logarithmic scaling, the interaction between the macro-world (R) and the micro-world (r) is governed by the invariant winding numbers of the torus. We define the fine-structure constant α as the pure, unscaled ratio of these spatial cycles, modified by the quadratic curvature factor $(4\pi^2)$ which represents the total surface embedding of the 2π-periodicity:

$$\alpha = \frac{2\pi r}{2\pi R \cdot 2\pi} = \frac{r}{2\pi R} \quad \Longrightarrow \quad \frac{R}{r} = \frac{1}{2\pi\alpha} \tag{4}$$

By transforming the mathematical framework from a logarithmic continuum to a projective modular ratio, the expansion of the universe becomes a linear, topological relaxation of the aspect ratio defined solely by the integer-bound winding properties of the space-time torus.

5 Resolution of the Hubble Tension and Local Gauge Invariance

The Hubble tension manifests as a statistically significant divergence between early-universe CMB measurements ($H_{\text{early}} \approx 67.4$ km/s/Mpc) [4] and late-universe local observations ($H_{\text{local}} \approx 73$ km/s/Mpc) [3]. We state that the physical expansion of the spatial metric is driven by the 4D accretion $\dot{M}_{4\text{D}}$, but the interpreted divergence—the tension—is an analytical projection artifact caused by forcing a variable vacuum resonance into a static, flat coordinate system.

We define the local speed of light as a dynamic gauge parameter:

$$c(t) = f_0 \cdot 2\pi R(t) = f_0 \cdot \frac{r(t)}{\alpha(t)} \tag{5}$$

To preserve local Lorentz invariance and remain compliant with strict terrestrial atomic clock experiments, we introduce a local conformal gauge transformation. Any local observer measuring c utilizes apparatuses bound by Bohr radii $a_0 = \frac{\hbar}{\alpha m_e c}$. As $c(t)$ and $\alpha(t)$ evolve adiabatically on cosmological scales, the local atomic spatial scales and mass energies shift proportionally:

$$\Delta\text{Local Measurement} \propto \frac{\dot{c}}{c} + \frac{\dot{\alpha}}{\alpha} = 0 \tag{6}$$

Consequently, c and α are measured as absolute constants in any local, comoving frame. The true discrepancy defining the Hubble tension (ΔH_0) is therefore equivalent to the exact drift rate of the vacuum resonance coupling over cosmic time:

$$\Delta H_0 \equiv H_{\text{local}} - H_{\text{early}} = -\frac{\dot{\alpha}}{\alpha} \tag{7}$$

By assigning the burden of the variable constants solely to the delta of the tension ($\Delta H_0 \approx 5.6$ km/s/Mpc) rather than the baseline expansion rate (H_0), the required drift is drastically reduced, bypassing prior empirical vetoes.

6 Exemplary Re-Calculation and Empirical Compliance

Converting the tension delta $\Delta H_0 = 5.6$ km/s/Mpc into SI base units yields

$\Delta H_0 \approx 1.81 \times 10^{-19}$ s^{-1}. Using our modified geometric relation, we solve for the real time-dependent variation of the fine-structure constant:

$$\dot{\alpha} = -\Delta H_0 \cdot \alpha \approx -1.32 \times 10^{-21} \text{ s}^{-1} \tag{8}$$

Aggregating this value over a terrestrial year ($\Delta t \approx 3.15 \times 10^7$ s), the relative annual variation of α translates to:

$$\frac{\Delta\alpha}{\alpha \cdot \text{year}} \approx 5.7 \times 10^{-14} \text{ yr}^{-1} \tag{9}$$

This drops safely near the threshold of modern astronomical bounds derived from distant quasar absorption spectra (10^{-15} to 10^{-16} yr^{-1}) [5] and remains utterly invisible to local atomic clock comparisons due to our conformal gauge coupling.

7 The Terminal Boundary: Vacuum Collapse and 4D Hypernova Feedback

Because the host 4D black hole continuously ingests matter ($\dot{M}_{4D} > 0$), the major radius $R(t)$ expands asymptotically. Consequently, according to Section 5, the fine-structure constant $\alpha(t)$ must rise continuously over cosmic time. This evolutionary trajectory dictates an absolute, inescapable terminal boundary for the 3D universe: the **Maximal Geometric Tension Point** at $\alpha \to 1$.

As α approaches unity, the fundamental equilibrium of nature collapses. The electromagnetic repulsion between subatomic particles intensifies quadratically, eventually matching and overriding the strong nuclear force. At this critical threshold, an abrupt **Vacuum Collapse** occurs: electrons fuse into protons, atomic structures disintegrate instantly, and the 3D space-time manifold loses its elastic structural stability.

Because information cannot be destroyed (Uni-Unitarity), this metric collapse triggers an instantaneous, cataclysmic energy feedback loop into the higher-dimensional bulk. The 3D event horizon "melts" due to information saturation. Denied its holographic boundary, the accumulated mass-energy stored over aeons within the 3D quantum vacuum flings back violently into the 4D bulk. For an observer in the 4D mother-universe, this causes the host black hole to lose its gravitational confinement and explode in a colossal **4D Hypernova**.

7.1 Empirical Corroboration: The Collapsar Analogy

While a global vacuum collapse is inherently unobservable from within the affected manifold, modern astrophysics provides a compelling, lower-dimensional empirical analogue to this feedback loop: the **Collapsar Model** of long-duration Gamma-Ray Bursts (GRBs) [6, 7].

In our 3D universe, when an ultra-massive Wolf-Rayet star undergoes core collapse, a newborn stellar-mass black hole is formed. The surrounding stellar envelopes immediately fall into the horizon at relativistic speeds. When this accretion rate exceeds a critical threshold, the black hole becomes "overfed", creating an extreme hydrodynamic and magnetic bottleneck.

Instead of silently swallowing the matter, the system triggers a violent feedback reaction: highly relativistic, collimated jets puncture the star, detonating it in a **Hypernova** and releasing a peak isotropic energy of up to 10^{52} ergs in a matter of seconds [8].

We propose that this exact mechanism occurs *mutatis mutandis* during the $\alpha \to 1$ transition. The 3D universe acts as the internal energy reservoir of the 4D black hole. When the spatial framework collapses, it mimics an extreme, instantaneous overfeeding event from the perspective of the 4D bulk. The resulting 4D hypernova explosion is the exact macroscopic manifestation of our 3D universe's terminal energy being recycled back into the higher-dimensional space to form new stellar progenitors, thereby closing the grand cyclic cascade.

8 Conclusion & Core Insights

By substituting linear arithmetic with a 2π-periodic Riemannian number torus embedded on a 4D black hole horizon, we achieve three major conceptual breakthroughs:

- **Causal Expansion without Singularities:** Cosmic expansion is driven by the physical mass accretion ($\dot{M}_{4D}$) of our host 4D black hole. The Big Bang was a smooth topological sheet transition, completely free of unphysical infinite densities.

- **Tension Elimination:** The Hubble tension is solved analytically as a mathematical projection error incurred by forcing a dynamic, curved metric into a static, flat $\mathbb{R}$-coordinate system.

- **The Cyclic Cascade:** The ultimate fate of our universe is an atomic vacuum collapse at $\alpha = 1$, culminating in a 4D hypernova explosion—mirrored by the physics of lower-dimensional collapsars—that recycles all cosmic information back into the higher-dimensional bulk.

- **Stochastic Eigenvalues:** The physical constants of our universe are determined by the initial, randomized 4D angular momentum (L_{4D}) of the ancestral star, organizing our local metric into a custom 3D accretion sphere.

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